

Layout Dependent Effect Setup Usage

Layout Dependent Effect

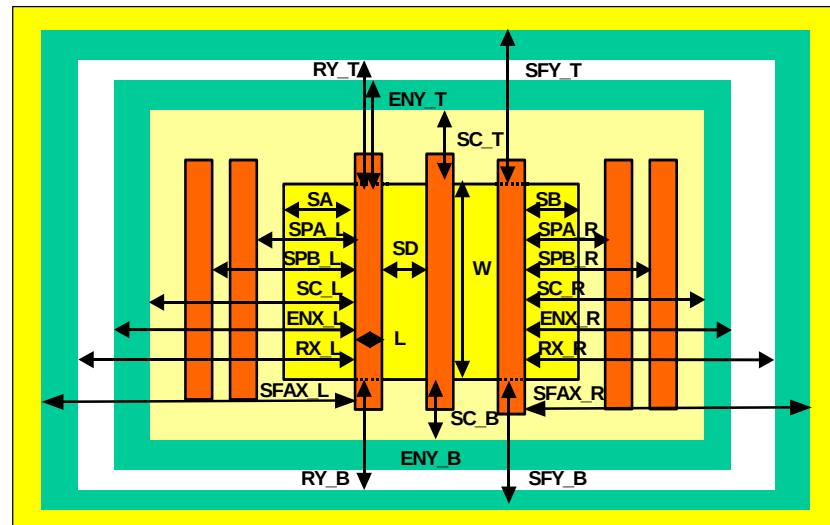
● Concept

- Define layout dependent effect (LDE) parameters by layer boundary and calculate effective parameters in pre-layout stage

● Method

- New adjustable parameters based on layout geometry are added into device CDF for layout dependent effect calculation
- User preset function is provided for customer's pre-defined values
- Utility is provided to allow user to specify those parameters by selected instance.

New CDF Parameters



* 20 new CDF parameters are introduced:

- (a) PSE: SPA_L, SPA_R, SPB_L, SPB_R
- (b) WPE: SC_L, SC_R, SC_T, SC_B
- (c) BE: ENX_L, ENX_R, ENY_T, ENY_B, RX_L, RX_R, RY_T, RY_B
- (d) OSE: SFAX_L, SFAX_R, SFY_T, SFY_B

OD
PO
WELL
CESL

WPE parameters

Add SC_L, SC_R, SC_T, SC_B CDF parameters

Edit Object Properties

Display S/D Parameters: ☒ OFF ☐ ON off

Display LOD Effect: ☒ OFF ☐ ON off

pre-set-Option: Preset1 off

Display Well Proximity Effect: ☐ OFF ☒ ON off

Well Proximity Effect Cal. Method: ☐ Model ☐ auto ☐ custom ☒ PreSet off

Gate_to_Left_NWell_Enc(SC_L) (M): 1u M off

Gate_to_Right_NWell_Enc(SC_R) (M): 2u M off

Gate_to_Top_NWell_Enc(SC_T) (M): 3u M off

Gate_to_Bottom_NWell_Enc(SC_B) (M): 4u M off

SCA: 1.38568 off

SCB: 3.97853e-05 off

SCC: 1.57089e-09 off

Display Poly Spacing Effect: ☐ OFF ☒ ON off

Poly Spacing Effect Cal. Method: ☐ Model ☐ auto ☐ custom ☒ PreSet off

Gate_Space_to_1st_Left_PO(SPA_L) (M): 5u M off

Gate_Space_to_1st_Right_PO(SPA_R) (M): 6u M off

Gate_Space_to_2nd_Left_PO(SPB_L) (M): 7u M off

Gate_Space_to_2nd_Right_PO(SPB_R) (M): 8u M off

SPA (M): 5.4569u M off

SPA1 (M): 5.43562u M off

SPA2 (M): 5.41025u M off

SPA3 (M): 5.43233u M off

SAP (M): 1.3091u M off

SAPB (M): 883.362n M off

OK Cancel Apply Defaults Previous Next Help

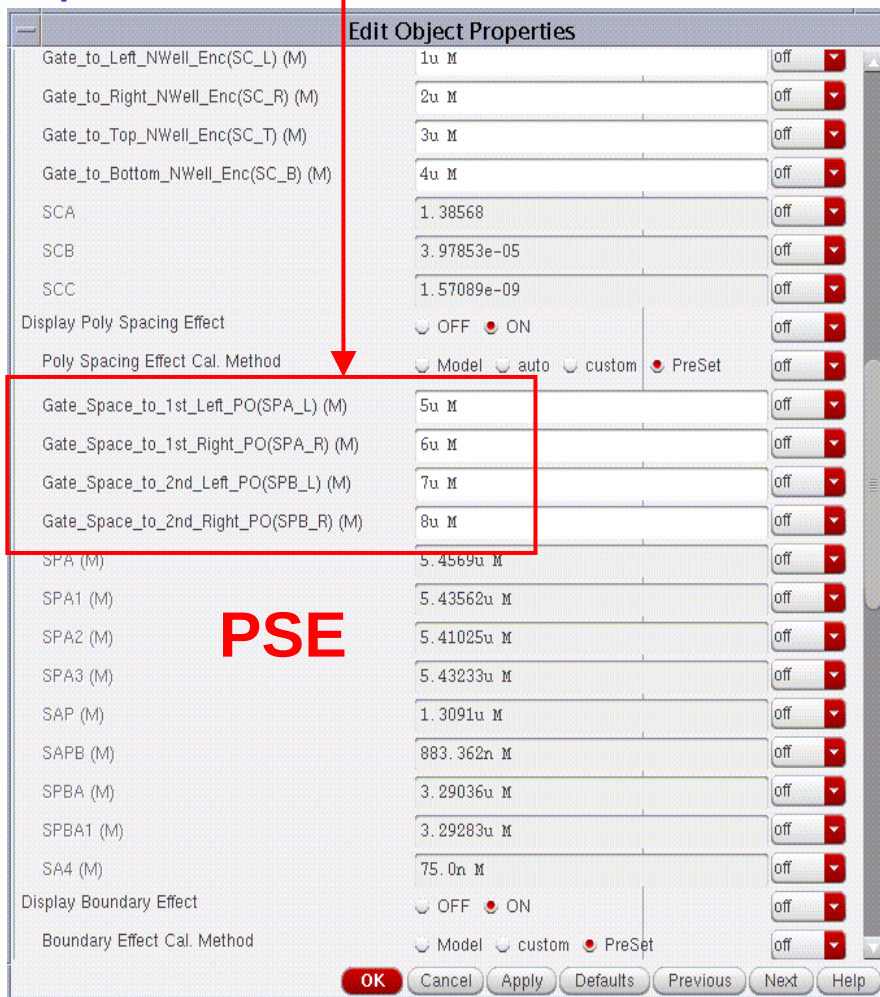
	New layout related input in CDF	layout dependent effect instance parameters
PSE	SPA_L, SPA_R, SPB_L, SPB_R	SPA, SPA1, SPA2, SPA3, SAP, SA4, SPBA, SPBA1, SAPB
WPE	SC_L, SC_R, SC_T, SC_B	SCA, SCB, SCC
BE (D-CESL)	ENX_L, ENX_R, ENY_T, ENY_B, RX_L, RX_R, RY_T, RY_B	ENX, ENX1, ENY, ENY1, ENY2, REX, REY
OSE	SFAX_L, SFAX_R, SFY_T, SFY_B	SODX, SODX1, SODX2, SA5, SA6, SODY

Calculate effective SCA, SCB, SCC

Ex : $SCA_{eff} = f(W, L, NF, SD, SC_L, SC_R, SC_T, SC_B)$

PSE parameters

Add SPA_L, SPA_R, SPB_L, SPB_R CDF parameters



PSE

Parameter	Value	Unit	Off/On
Gate_to_Left_NWell_Enc(SC_L) (M)	1u	M	off
Gate_to_Right_NWell_Enc(SC_R) (M)	2u	M	off
Gate_to_Top_NWell_Enc(SC_T) (M)	3u	M	off
Gate_to_Bottom_NWell_Enc(SC_B) (M)	4u	M	off
SCA	1.38568		off
SCB	3.97853e-05		off
SCC	1.57089e-09		off
Display Poly Spacing Effect	☐ OFF ☒ ON		off
Poly Spacing Effect Cal. Method	☐ Model ☐ auto ☐ custom ☒ PreSet		off
Gate_Space_to_1st_Left_PO(SPA_L) (M)	5u	M	off
Gate_Space_to_1st_Right_PO(SPA_R) (M)	6u	M	off
Gate_Space_to_2nd_Left_PO(SPB_L) (M)	7u	M	off
Gate_Space_to_2nd_Right_PO(SPB_R) (M)	8u	M	off
SPA (M)	5.4569u	M	off
SPA1 (M)	5.43562u	M	off
SPA2 (M)	5.41025u	M	off
SPA3 (M)	5.43233u	M	off
SAP (M)	1.3091u	M	off
SAPB (M)	883.362n	M	off
SPBA (M)	3.29036u	M	off
SPBA1 (M)	3.29283u	M	off
SA4 (M)	75.0n	M	off
Display Boundary Effect	☐ OFF ☒ ON		off
Boundary Effect Cal. Method	☐ Model ☐ custom ☒ PreSet		off

OK Cancel Apply Defaults Previous Next Help

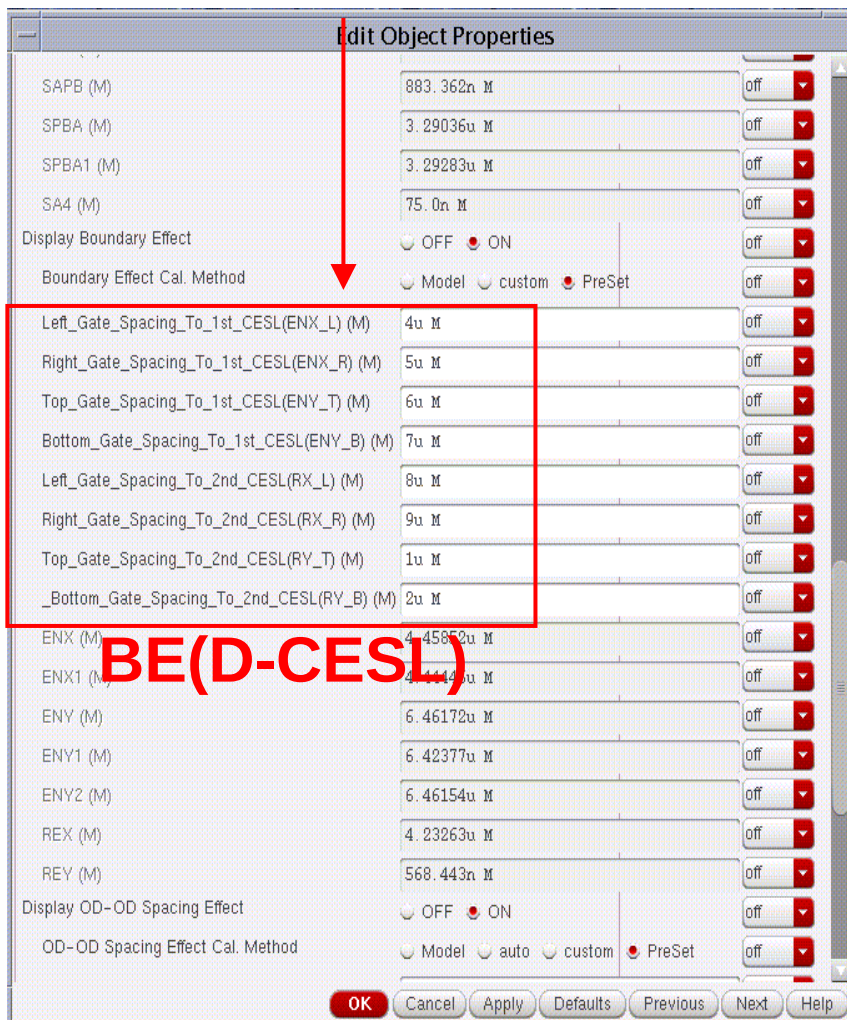
	New layout related input in CDF	layout dependent effect instance parameters
PSE	SPA_L, SPA_R, SPB_L, SPB_R	SPA, SPA1, SPA2, SPA3, SAP, SA4, SPBA, SPBA1, SAPB
WPE	SC_L, SC_R, SC_T, SC_B	SCA, SCB, SCC
BE (D-CESL)	ENX_L, ENX_R, ENY_T, ENY_B, RX_L, RX_R, RY_T, RY_B	ENX, ENX1, ENY, ENY1, ENY2, REX, REY
OSE	SFAX_L, SFAX_R, SFY_T, SFY_B	SODX, SODX1, SODX2, SA5, SA6, SODY

- PSE (Poly Spacing Effect) parameters are adjustable in CDF for callback calculation

Boundary Effect parameters

Add ENX_L, ENX_R, ENY_T, ENY_B,

RX_L, RX_R, RY_T, RY_B CDF parameters



BE(D-CESL)

	New layout related input in CDF	layout dependent effect instance parameters
PSE	SPA_L, SPA_R, SPB_L, SPB_R	SPA, SPA1, SPA2, SPA3, SAP, SA4, SPBA, SPBA1, SAPB
WPE	SC_L, SC_R, SC_T, SC_B	SCA, SCB, SCC
BE (D-CESL)	ENX_L, ENX_R, ENY_T, ENY_B, RX_L, RX_R, RY_T, RY_B	ENX, ENX1, ENY, ENY1, ENY2, REX, REY
OSE	SPAX_L, SPAX_R, SFY_T, SFY_B	SODX, SODX1, SODX2, SA5, SA6, SODY

- BE (D-CESL Effect) parameters are adjustable in CDF for callback calculation

OSE parameters

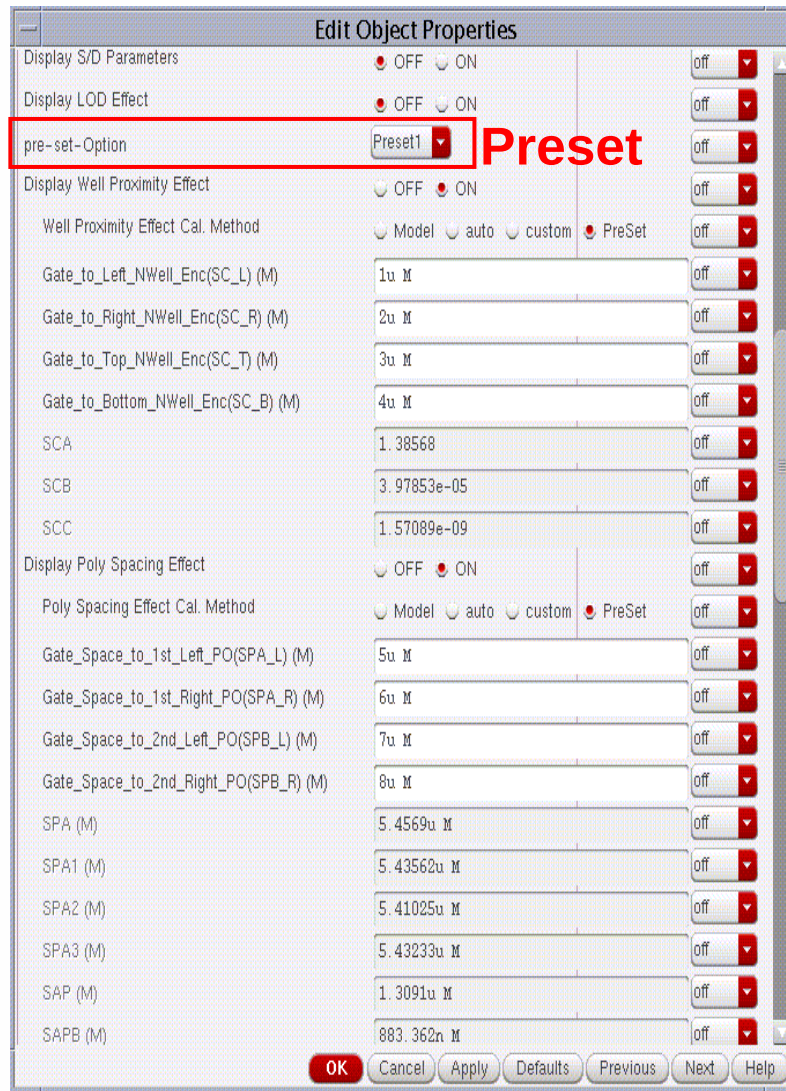
Add SFAX_L, SFAX_R, SFY_T, SFY_B CDF parameters

OSE

	New layout related input in CDF	layout dependent effect instance parameters
PSE	SPA_L, SPA_R, SPB_L, SPB_R	SPA, SPA1, SPA2, SPA3, SAP, SA4, SPBA, SPBA1, SAPB
WPE	SC_L, SC_R, SC_T, SC_B	SCA, SCB, SCC
BE (D-CESL)	ENX_L, ENX_R, ENY_T, ENY_B, RX_L, RX_R, RY_T, RY_B	ENX, ENX1, ENY, ENY1, ENY2, REX, REY
OSE	SFAX_L, SFAX_R, SFY_T, SFY_B	SODX, SODX1, SODX2, SA5, SA6, SODY

- OSE (OD Spacing Effect) parameters are adjustable in CDF for callback calculation

Preset Option



The 'Edit Object Properties' dialog box shows various parameters for a layout object. The 'pre-set-Option' is highlighted with a red box and labeled 'Preset'. The 'Well Proximity Effect Cal. Method' is set to 'PreSet'. The 'Poly Spacing Effect' is also set to 'PreSet'.

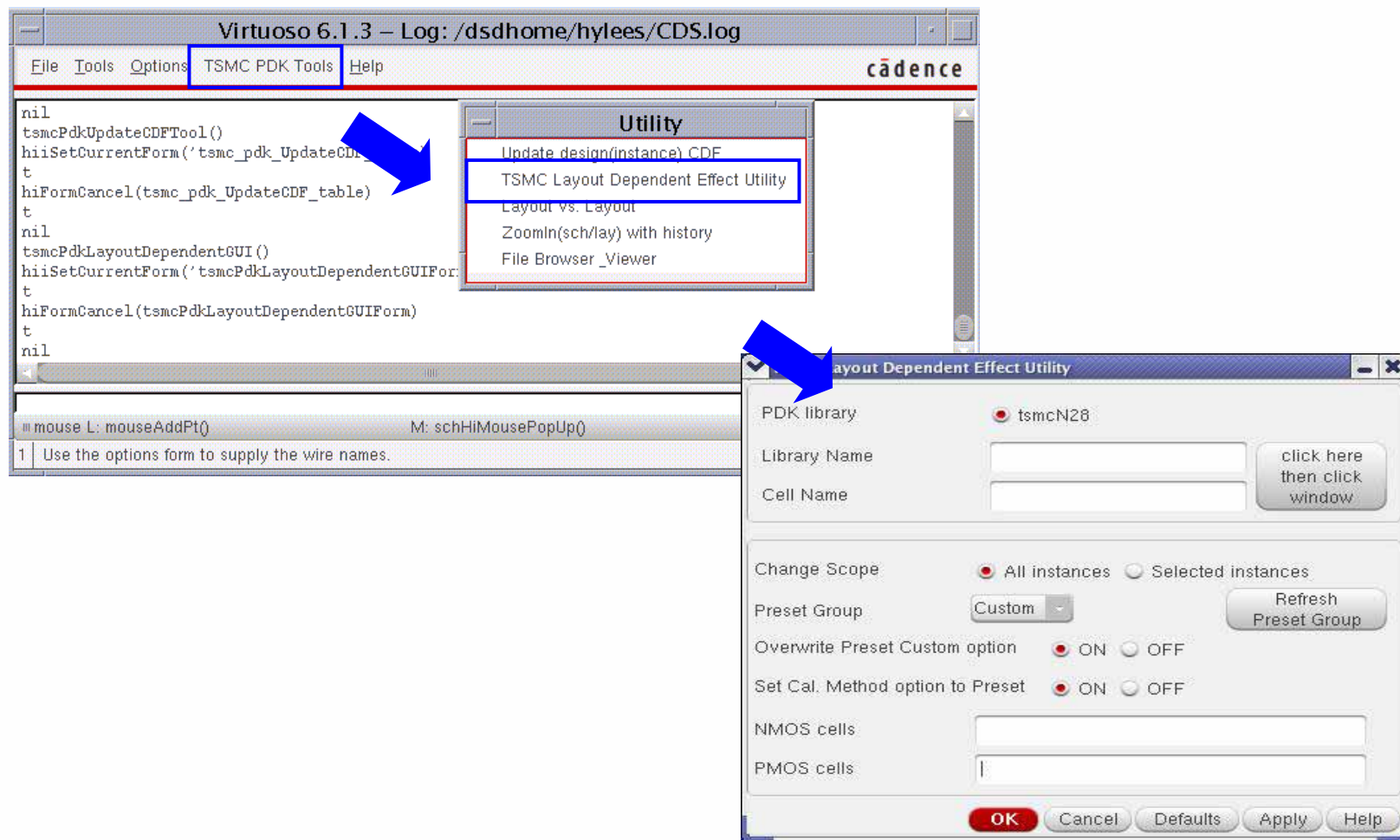
- Preset function is provided for customer's pre-defined parameters
- TSMC utility is provided for user selecting different groups of preset.

I	/sk111/TSMC_LayoutEffect	IW	TSMC_LayoutEffectTable
[Preset1]		[Preset2]	
<NMOS>		<NMOS>	
<wpe>		<wpe>	
SC_L = 1u		SA = 140n	
SC_R = 2u		SB = 140n	
SC_T = 3u		</tod>	
SC_B = 4u		<wpe>	
</wpe>		SC_L = 1u	
<pse>		SC_R = 1u	
SPA_L = 5u		SC_T = 1u	
SPA_R = 6u		SC_B = 1u	
SPB_L = 7u		</wpe>	
SPB_R = 8u		<pse>	
</pse>		SPA_L = 0.17u	
<ose>		SPA_R = 0.17u	
SFAX_L = 9u		SPB_L = 4u	
SFAX_R = 1u		SPB_R = 4u	
SFY_T = 2u		</pse>	
SFY_B = 3u		<ose>	
</ose>		SFAX_L = 0.24u	
<ces1>		SFAX_R = 0.24u	
ENX_L = 4u		SFY_T = 0.88u	
ENX_R = 5u		SFY_B = 0.88u	
ENY_T = 6u		</ose>	
ENY_B = 7u		<ces1>	
RX_L = 8u		ENX_L = 4u	
RX_R = 9u		ENX_R = 5u	
RY_T = 1u		ENY_T = 6u	
RY_B = 2u		ENY_B = 7u	
</ces1>		RX_L = 8u	
</NMOS>		RX_R = 9u	
<PMOS>		RY_T = 1u	
<wpe>		RY_B = 2u	
SC_L = 1u		</ces1>	
SC_R = 2u		</NMOS>	
SC_T = 3u		<PMOS>	
SC_B = 4u		<Tod>	
</wpe>		SA = 110n	
<pse>		SB = 110n	
SPA_L = 5u		</tod>	
SPA_R = 6u		<wpe>	
SPB_L = 7u		SC_L = 1u	
SPB_R = 8u		SC_R = 1u	
</pse>		SC_T = 1u	
		SC_B = 1u	
		</wpe>	
		<pse>	
		SPA_L = 0.14u	
		SPA_R = 0.14u	
		SPB_L = 4u	
		SPB_R = 4u	

Preset1 table

Preset2 table

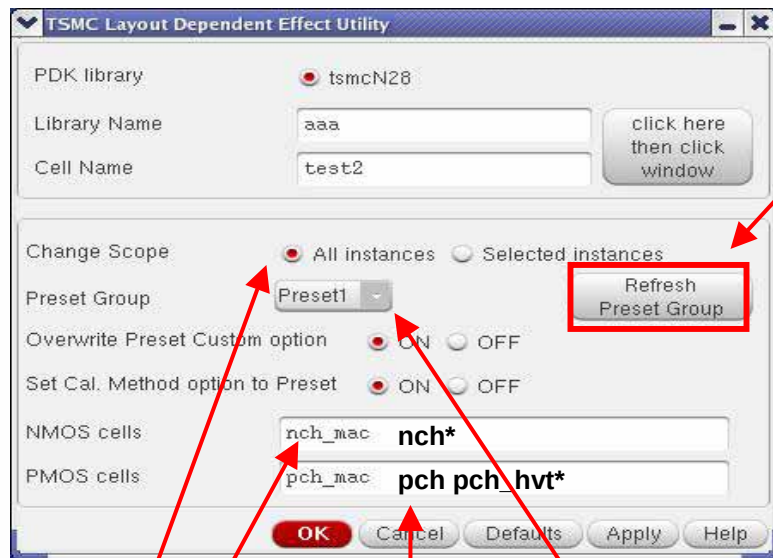
TSMC PDK Tools-Layout Dependent Effect Utility



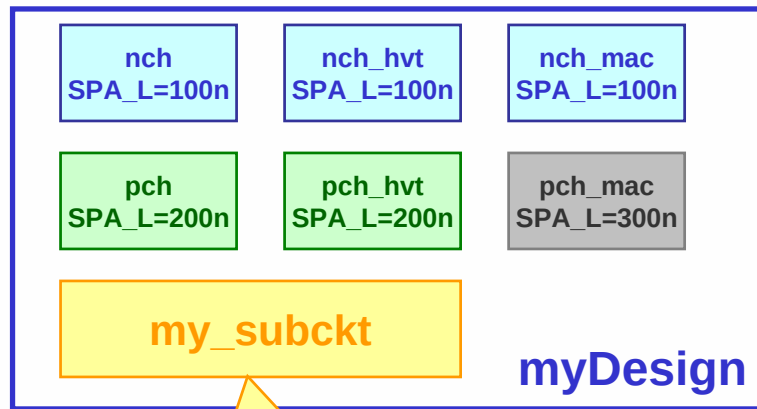
The screenshot shows the Virtuoso 6.1.3 interface with the TSMC PDK Tools menu open. A blue arrow points to the 'TSMC Layout Dependent Effect Utility' option. Another blue arrow points to the 'Layout Dependent Effect Utility' dialog box, which is open. The dialog box contains the following fields and options:

- PDK library:** tsmcN28
- Library Name:** (empty field)
- Cell Name:** (empty field)
- Change Scope:** ☒ All instances ☐ Selected instances
- Preset Group:** Custom (dropdown menu)
- Overwrite Preset Custom option:** ☒ ON ☐ OFF
- Set Cal. Method option to Preset:** ☒ ON ☐ OFF
- NMOS cells:** (empty field)
- PMOS cells:** (empty field)
- Buttons:** OK, Cancel, Defaults, Apply, Help

Specification of Layout Dependent Effect Utility



Press the button to refresh your preset group



Support wildcard and cell list (separated by space)

SPA_L=100n will be applied to "All instances" & "nch*" in cell "myDesign". Which include nch, nch_hvt, nch_mac and my_subckt/nch

SPA_L=200n will be applied to "All instances" & (pch || pch_hvt) in cell "myDesign". Which include pch, pch_hvt and my_subckt/pch.

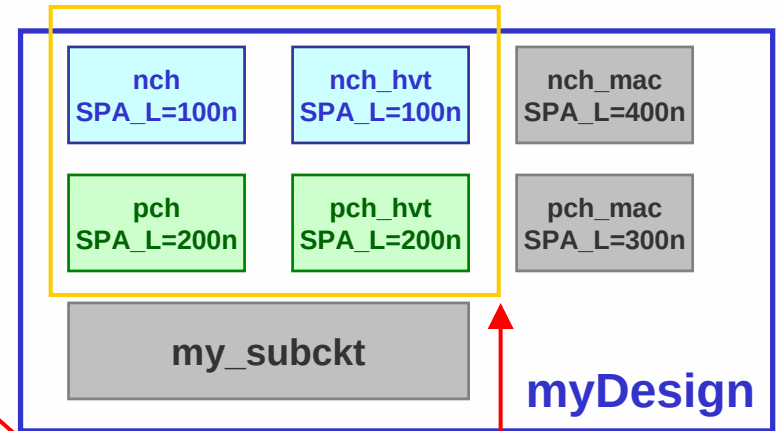
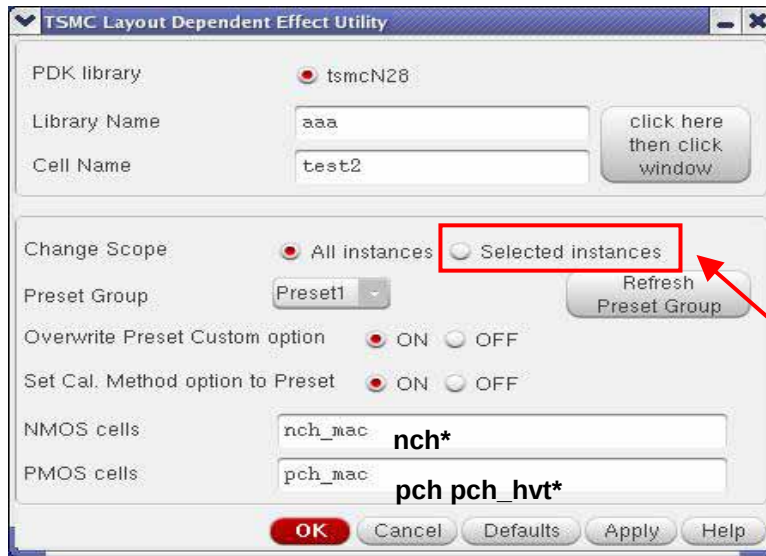
Control File

```
<Preset 1>
NMOS
SPA_L = 100n
SPA_R = xxx
...
PMOS
SPA_L = 200n
SPA_R = xxx
...
<Preset 2>
...
```

The utility also support hierarchical instances (for more than one level).

**** The utility can only affect those instances with write permission.**

Specification of Layout Dependent Effect Utility (Cont.)



Only apply the change to those instances that selected by mouse.

SPA_L=100n will be applied to "Selected instances" && "nch*" in cell "myDesign". Which include nch, nch_hvt

SPA_L=200n will be applied to "Selected instances" && (pch || pch_hvt) in cell "myDesign". Which include pch, pch_hvt .

Control File

```
<Preset 1>
NMOS
  SPA_L = 100n
  SPA_R = xxx
  ...
PMOS
  SPA_L = 200n
  SPA_R = xxx
  ...
<Preset 2>
...
```

The control file is put at the skill directory named "TSMC_LayoutEffectTable". Hint, there are no limit for preset group.

Specification of Layout Dependent Effect Utility (Cont.)

TSMC Layout Dependent Effect Utility

PDK library: ☒ tsmcN28

Library Name: aaa

Cell Name: test2

Change Scope: ☒ All instances ☐ Selected instances

Preset Group: Preset1

Overwrite Preset Custom option: ☒ ON ☐ OFF

Set Cal. Method option to Preset: ☒ ON ☐ OFF

NMOS cells: nch_mac nch*

PMOS cells: pch_mac pch_hvt*

OK Cancel Defaults Apply Help

Enable to overwrite the preset option when the original preset option of a instance is "Custom"

For example, When you turn off this option, This utility will not change the following instance since Its preset option is "Custom"

Enable to reset "Cal. Method option" to Preset

For example, When you turn on this option, This utility will change all Cal. Method CDF options to "Preset".

Display LOD Effect: ☐ OFF ☐ ON

Previous pre-set: Preset2

pre-set-Option: Custom

Display Well Proximity Effect: ☐ OFF ☒ ON

Well Proximity Effect Cal. Method: ☐ Model ☐ auto ☐ custom ☒ PreSet

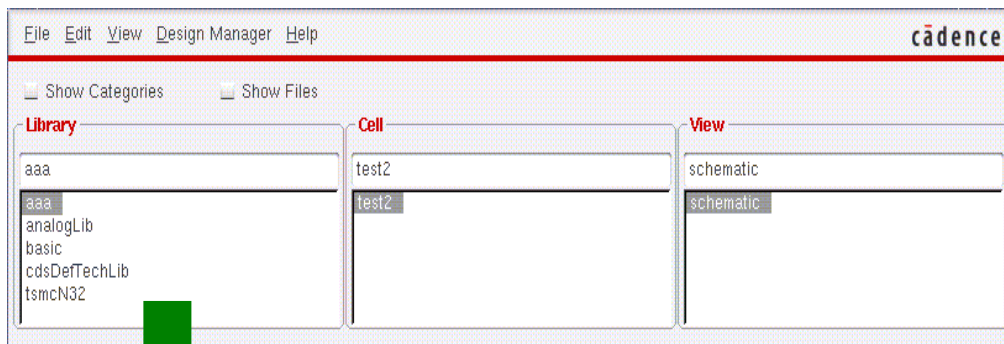
Gate_to_Left_NWell_Enc(SC_L) (M): 1u M

Gate_to_Right_NWell_Enc(SC_R) (M): 1u M

Gate_to_Top_NWell_Enc(SC_T) (M): 2u M

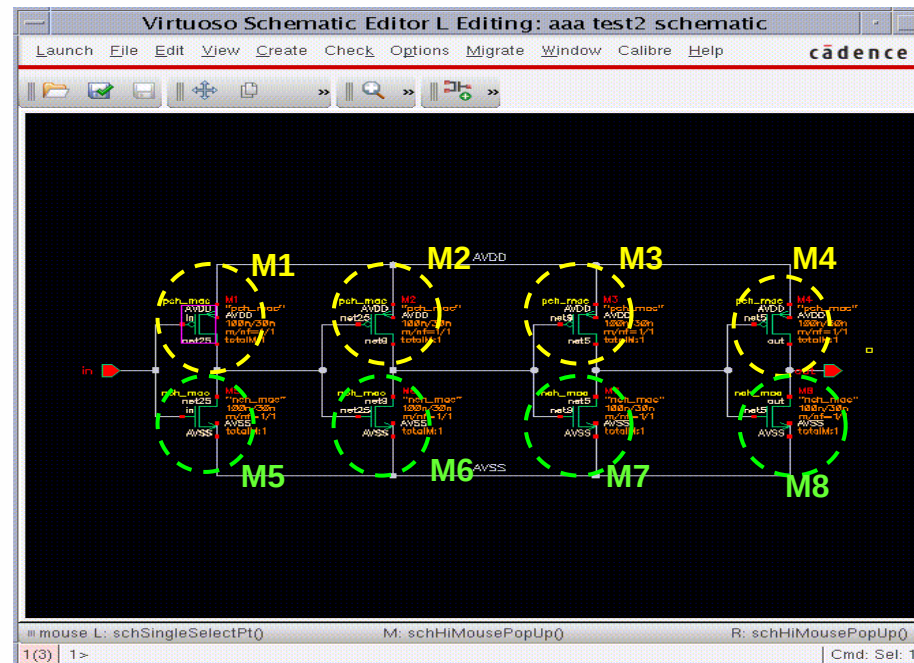
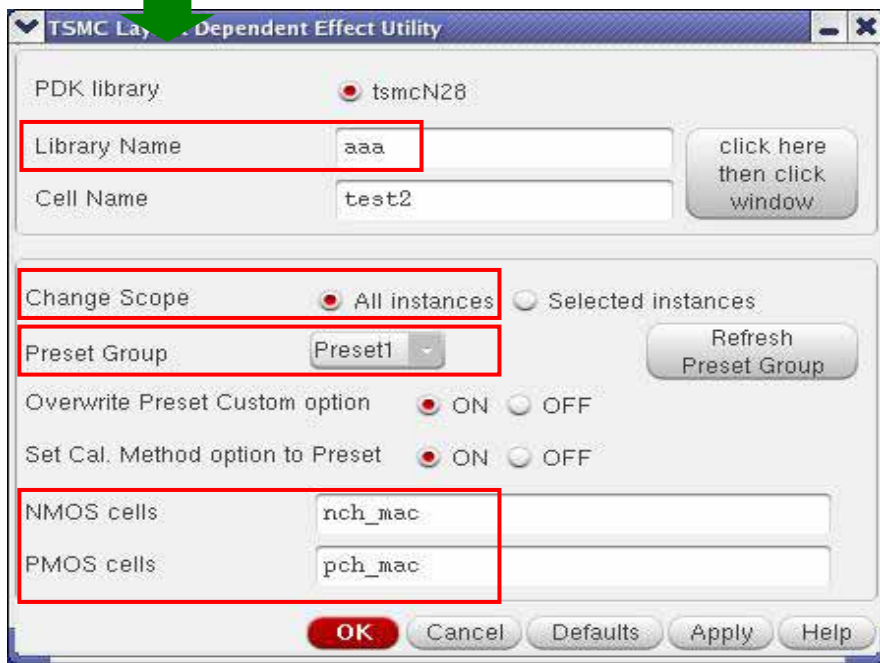
Gate_to_Bottom_NWell_Enc(SC_B) (M): 1u M

Example 1: Setting LDE parameters of driver by “All instances”



NMOS LDE parameters set to “Preset1”
(M1~M4)

PMOS LDE parameters set to “Preset1”
(M5~M8)



Example 1: Setting LDE parameters of driver by "All instances" (Cont.)

Virtuoso Schematic Editor L Editing: aaa test2 schematic

Launch File Edit View Create Check Options Migrate Window Calibre Help cadence

CDF of M4

Edit Object Properties

Value	Display
pch_mac	off
0.85V standard VT PMOS transistor	off
30n M	off
100n M	off
100n M	off
1	off
1	off
1	off
☒ OFF ☐ ON	off
☐ OFF ☐ ON	off
☐ OFF ☐ ON	off
Preset1	off
☐ Model ☐ auto ☐ custom ☒ PreSet	off
Gate_to_Left_NWell_Enc(SC_L) (M)	1u M
Gate_to_Right_NWell_Enc(SC_R) (M)	2u M
Gate_to_Top_NWell_Enc(SC_T) (M)	3u M
Gate_to_Bottom_NWell_Enc(SC_B) (M)	4u M
SCA	1.38568
SCB	3.97853e-05
SCC	1.57089e-09
☐ OFF ☐ ON	off

pre-set-Option
Display Well Proximity Effect
Well Proximity Effect Cal. Method

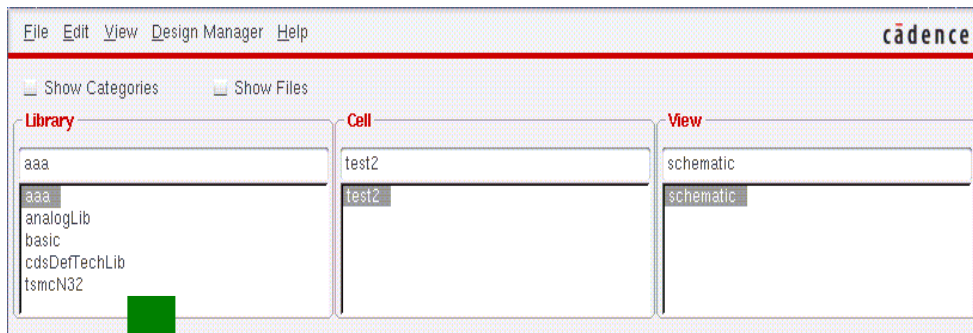
SCA = 1.38568
SCB = 3.97853e-05
SCC = 1.57089e-09

```

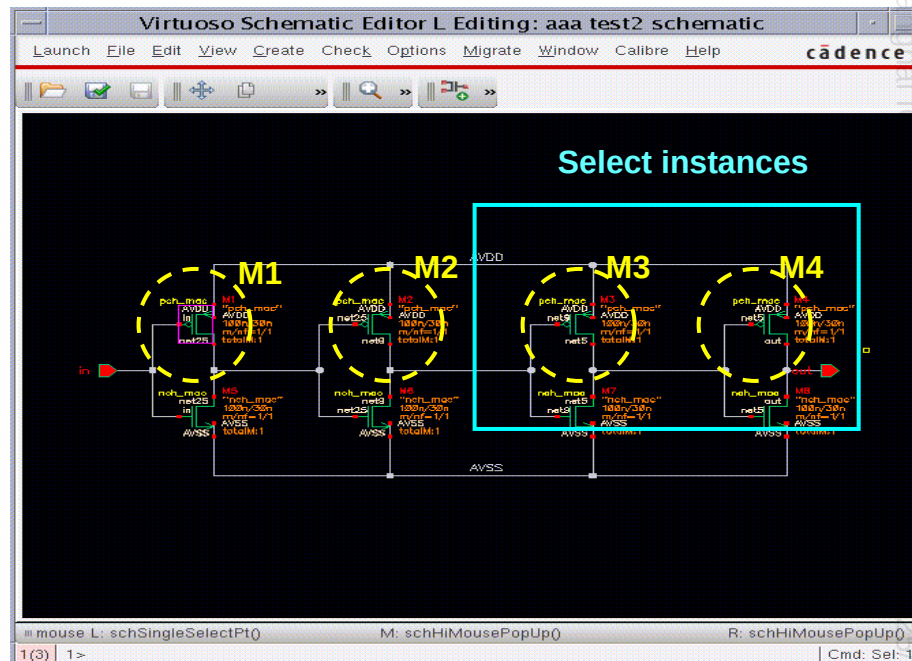
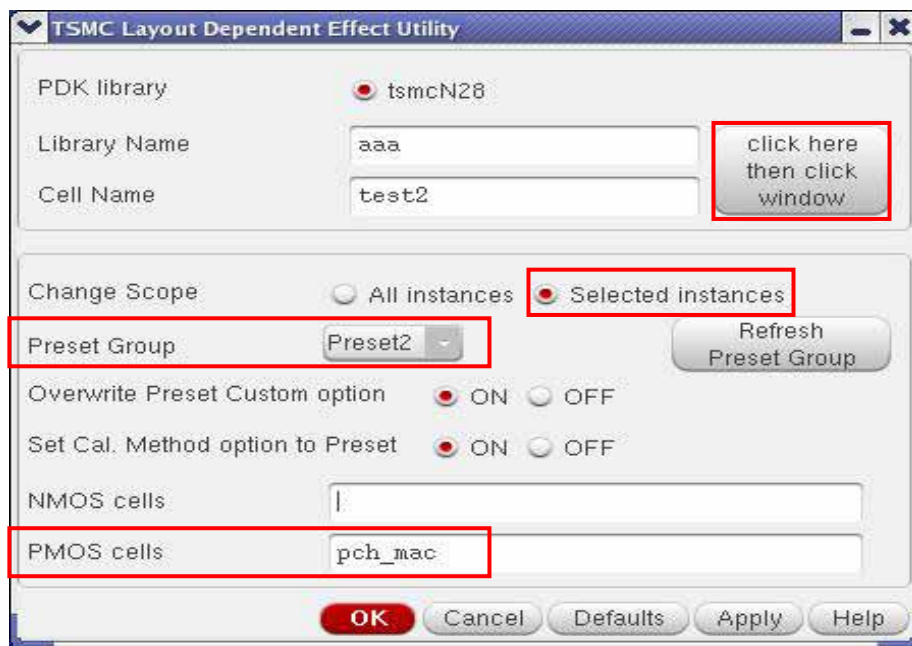
I ./skill/TSMC_LayoutEffect
[Preset1]
<NMOS>
  <wpe>
    SC_L = 1u
    SC_R = 2u
    SC_T = 3u
    SC_B = 4u
  </wpe>
  <pse>
    SPA_L = 5u
    SPA_R = 6u
    SPB_L = 7u
    SPB_R = 8u
  </pse>
  <ose>
    SFAX_L = 9u
    SFAX_R = 1u
    SFY_T = 2u
    SFY_B = 3u
  </ose>
  <ces1>
    ENX_L = 4u
    ENX_R = 5u
    ENY_T = 6u
    ENY_B = 7u
    RX_L = 8u
    RX_R = 9u
    RY_T = 1u
    RY_B = 2u
  </ces1>
</NMOS>
<PMOS>
  <wpe>
    SC_L = 1u
    SC_R = 2u
    SC_T = 3u
    SC_B = 4u
  </wpe>
  <pse>
    SPA_L = 5u
    SPA_R = 6u
    SPB_L = 7u
    SPB_R = 8u
  </pse>
  
```

OK Cancel Apply Defaults Previous Next Help

Example 2: Setting LDE parameters of driver by “Select instances”



PMOS LDE parameters set to “Preset2”
(M3~M4)



Example 2: Setting LDE parameters of driver by "Select instances" (Cont.)

CDF of M3/M4

Virtuoso Schematic Editor L Editing: aaa test2 schematic

Object Properties

Display

pch_mac

0.85V standard VT PMOS transistor

30n M

100n M

100n M

1

1

1

OFF ON

OFF ON

OFF ON

Preset2

Model auto cut Sim PreSet

Gate_to_Left_NWell_Enc(SC_L) (M)

Gate_to_Right_NWell_Enc(SC_R) (M)

Gate_to_Top_NWell_Enc(SC_T) (M)

Gate_to_Bottom_NWell_Enc(SC_B) (M)

SCA

SCB

SCC

Display Poly Spacing Effect

OFF ON

OK Cancel Apply Defaults Previous Next Help

IW TSMC_LayoutEffectTable

[Preset2]

<NMOS>

<1od>

SA = 140n

SB = 140n

</1od>

<wpe>

SC_L = 1u

SC_R = 1u

SC_T = 1u

SC_B = 1u

</wpe>

<pse>

SPA_L = 0.17u

SPA_R = 0.17u

SPB_L = 4u

SPB_R = 4u

</pse>

<ose>

SFAX_L = 0.24u

SFAX_R = 0.24u

SFY_T = 0.88u

SFY_B = 0.88u

</ose>

<ces1>

ENX_L = 4u

ENX_R = 5u

ENY_T = 6u

ENY_B = 7u

RX_L = 8u

RX_R = 9u

RY_T = 1u

RY_B = 2u

</ces1>

</NMOS>

<PMOS>

<1od>

SA = 110n

SB = 110n

</1od>

<wpe>

SC_L = 1u

SC_R = 1u

SC_T = 1u

SC_B = 1u

</wpe>

<pse>

SPA_L = 0.14u

SPA_R = 0.14u

SPB_L = 4u

SPB_R = 4u

SCA =3.75993

SCB =0.000139359

SCC =4.98519e-09

Example 3: Record the history of modification

Step1:TSMC preset LDE table of "Preset2"

Step2:Modify SC_T=1u => 2u

```

IW TSMC_LayoutEffectTable
[Preset2]
<NMOS>
<Tod>
  SA = 140n
  SB = 140n
</Tod>
<wpe>
  SC_L = 1u
  SC_R = 1u
  SC_T = 1u
  SC_B = 1u
</wpe>
<pse>
  SPA_L = 0.17u
  SPA_R = 0.17u
  SPB_L = 4u
  SPB_R = 4u
</pse>
<ose>
  SFAX_L = 0.24u
  SFAX_R = 0.24u
  SFY_T = 0.88u
  SFY_B = 0.88u
</ose>
<ces1>
  ENX_L = 4u
  ENX_R = 5u
  ENY_T = 6u
  ENY_B = 7u
  RX_L = 8u
  RX_R = 9u
  RY_T = 1u
  RY_B = 2u
</ces1>
</NMOS>
<PMOS>
<Tod>
  SA = 110n
  SB = 110n
</Tod>
<wpe>
  SC_L = 1u
  SC_R = 1u
  SC_T = 1u
  SC_B = 1u
</wpe>
<pse>
  SPA_L = 0.14u
  SPA_R = 0.14u
  SPB_L = 4u
  SPB_R = 4u
  
```

Step1

Model name	Value	Display
Model name	pcn_mac	off
description	0.85V standard VT PMOS transistor	off
Length(M)	30n M	off
Finger_width(M)	100n M	off
Width(M)	100n M	off
Number of Fingers	1	off
Multiplier	1	off
Hard_constrain	☒	off
Display Simulation Flag	☐ OFF ☐ ON	off
Display S/D Parameters	☐ OFF ☐ ON	off
Display LOD Effect	☐ OFF ☐ ON	off
pre-set-Option	Preset2	off
Display Well Proximity Effect	☐ OFF ☒ ON	off
Well Proximity Effect Cal. Method	☐ Model ☐ auto ☐ custom ☒ PreSet	off
Gate_to_Left_NWell_Enc(SC_L) (M)	1u M	off
Gate_to_Right_NWell_Enc(SC_R) (M)	1u M	off
Gate_to_Top_NWell_Enc(SC_T) (M)	1u M	off
Gate_to_Bottom_NWell_Enc(SC_B) (M)	1u M	off
SCA	3.75993	off
SCB	0.000139359	off
SCC	4.98519e-09	off
Display Poly Spacing Effect	☐ OFF ☐ ON	off

Step2

Model name	Value	Display
Model name	pcn_mac	off
description	0.85V standard VT PMOS transistor	off
Length(M)	30n M	off
Finger_width(M)	100n M	off
Width(M)	100n M	off
Number of Fingers	1	off
Multiplier	1	off
Hard_constrain	☒	off
Display Simulation Flag	☐ OFF ☐ ON	off
Display S/D Parameters	☐ OFF ☐ ON	off
Display LOD Effect	☐ OFF ☐ ON	off
Previous pre-set	Preset2	off
pre-set-Option	Custom	off
Display Well Proximity Effect	☐ OFF ☒ ON	off
Well Proximity Effect Cal. Method	☐ Model ☐ auto ☐ custom ☒ PreSet	off
Gate_to_Left_NWell_Enc(SC_L) (M)	1u M	off
Gate_to_Right_NWell_Enc(SC_R) (M)	1u M	off
Gate_to_Top_NWell_Enc(SC_T) (M)	2u M	off
Gate_to_Bottom_NWell_Enc(SC_B) (M)	1u M	off
SCA	3.08893	off
SCB	0.000109464	off
SCC	4.06348e-09	off
Display Poly Spacing Effect	☐ OFF ☐ ON	off

● Users still can change LDE parameters
For example of "SC_T"

● User modify "SC_T to 2um"

● Get new effective WPE parameters
SCA =3.08893
SCB =0.000109464
SCC =4.06348e-09

Example 3: Record the history of modification tsmc (Cont.)

Proprietary
& Confidential

Step3: CDF shows **pre-set-Option** has been modified by custom

Step4: CDF records original LDE parameters from "**Preset2**"

CDF Parameter	Value	Display
Model name	pch_mac	off
description	0.85V standard VT PMOS transistor	off
Length(M)	30n M	off
Finger_width(M)	100n M	off
Width(M)	100n M	off
Number of Fingers	1	off
Multiplier	1	off
total_m	1	off
Hard_constraint	☒	off
Display Simulation Flag	☐ OFF ☐ ON	off
Display S/D Parameters	☐ OFF ☐ ON	off
Display LOD Effect	☐ OFF ☐ ON	off
pre-set-Option	Preset2	off
Display Well Proximity Effect	☐ OFF ☒ ON	off
Well Proximity Effect Cal. Method	☐ Model ☐ auto ☐ custom ☒ PreSet	off
Gate_to_Left_NWell_Enc(SC_L) (M)	1u M	off
Gate_to_Right_NWell_Enc(SC_R) (M)	1u M	off
Gate_to_Top_NWell_Enc(SC_T) (M)	1u M	off
Gate_to_Bottom_NWell_Enc(SC_B) (M)	1u M	off
SCA	3.75993	off
SCB	0.000139359	off
SCC	4.98519e-09	off
Display Poly Spacing Effect	☐ OFF ☐ ON	off

Step3-4

CDF Parameter	Value	Display
Model name	pch_mac	off
description	0.85V standard VT PMOS transistor	off
Length(M)	30n M	off
Finger_width(M)	100n M	off
Width(M)	100n M	off
Number of Fingers	1	off
Multiplier	1	off
total_m	1	off
Hard_constraint	☒	off
Display Simulation Flag	☐ OFF ☐ ON	off
Display S/D Parameters	☐ OFF ☐ ON	off
Display LOD Effect	☐ OFF ☐ ON	off
Previous pre-set	Preset2	off
pre-set-Option	Custom	off
Display Well Proximity Effect	☐ OFF ☒ ON	off
Well Proximity Effect Cal. Method	☐ Model ☐ auto ☐ custom ☒ PreSet	off
Gate_to_Left_NWell_Enc(SC_L) (M)	1u M	off
Gate_to_Right_NWell_Enc(SC_R) (M)	1u M	off
Gate_to_Top_NWell_Enc(SC_T) (M)	2u M	off
Gate_to_Bottom_NWell_Enc(SC_B) (M)	1u M	off
SCA	3.08893	off
SCB	0.000109464	off
SCC	4.06348e-09	off
Display Poly Spacing Effect	☐ OFF ☐ ON	off

Now CDF shows "**Custom**" modify

Record the previous "**pre-set**" table from **Preset2**